

Chapter 5

From Macro to Micro Architecture

5.1 Overview

Wearable devices now stage sleep, estimate heart rate from light passed through the skin, and remove motion noise as it happens, all continuously on a device the size of a watch face. Blood pressure, one of the most basic hemodynamic measurements, is still taken much as it was around 1900. A cuff is inflated around the upper arm until it stops the flow of blood, then slowly released while the device listens for the sounds, or senses the pressure oscillations, of blood forcing its way back through the compressed artery. The result is a single reading of a quantity that changes with every heartbeat.

The continuous reference is invasive. A catheter carrying a pressure sensor is placed inside an artery, which is done only in intensive care and during surgery. Neither the cuff nor the catheter allows continuous monitoring in daily life.

Pulse transit time offers a way around both. Each heartbeat launches a pressure wave along the arteries. Timing that wave between two points on the body gives its speed, and the wave travels faster when pressure is higher. In principle, blood pressure then follows from a timing measurement and a calibration.

In practice the artery is not a passive pipe. When the left ventricle ejects blood into the aorta, the blood itself moves at about 1 m/s at peak systole. The ejection also stretches the arterial wall, and this stretch travels along the arterial tree at 4 to 15 m/s, depending on how stiff the arteries are. Transit time measures this wave in the wall, not the movement of the blood, and the stiffness of the wall depends on pressure in a way that differs between people (Section 5.2).

Chapter 3 constrained what a model sees. Chapter 4 used blood flow principles to constrain the macro architecture of a model, at the level of cross-attention between measurement sites. This chapter moves to the computation inside the model and asks whether it can be verified. Blood pressure is the test case because it is the hemodynamic quantity for which an unverified model is most dangerous. A cuffless estimator can match reference pressure on a benchmark while computing nothing related to pulse propagation, and accuracy alone cannot reveal this.

The chapter has two parts. Project 3A (Section 5.3) is a mechanistic audit that tests whether an estimator’s prediction depends on transit time in the way the physics requires, with preliminary results on a synthetic testbed. Project 3B (Section 5.4) applies a second set of blood flow principles, from how the brain keeps and prunes its vessels, to the micro architecture of a transformer. A rule we call hemodynamic attenuation closes attention heads during training

until only the circuit the task needs remains, so that the computation which survives training is small enough to audit.

5.2 Background

5.2.1 Transit time and pressure

The speed of a pressure wave in an elastic tube is given by the Moens–Korteweg relation, $PWV = \sqrt{Eh/(\rho d)}$, where E is the elastic modulus of the wall, h its thickness, d the vessel diameter and ρ the density of blood. The wall stiffens as pressure rises, which is commonly modelled as $E = E_0 e^{\zeta P}$ [1]. Combining the two gives

$$PWV = \sqrt{\frac{E_0 h}{\rho d}} e^{\zeta P/2}, \quad P = \frac{2}{\zeta} \ln PWV - \frac{1}{\zeta} \ln \frac{E_0 h}{\rho d}. \quad (5.1)$$

Pressure is a logarithmic function of wave speed with constants that differ between people, which is why transit-time methods need calibration [2].

5.2.2 Arrival time is not transit time

Most cuffless systems measure pulse arrival time (PAT), from the R-peak of the ECG to the foot of the pulse in a peripheral PPG. This interval also contains the pre-ejection period, the delay between the electrical activation of the heart and the opening of the aortic valve:

$$PAT = PEP + PTT. \quad (5.2)$$

PEP changes with contractility and sympathetic tone independently of arterial pressure, so any model trained on PAT inherits this confound. For a site at path length d_i from the heart, $PAT_i = PEP + d_i/PWV$, so the difference between two sites cancels PEP. Deep networks trained on raw ECG and PPG inherit the confound without showing it, and may also use regression toward the population mean, demographic priors, or waveform shape unrelated to propagation.

5.2.3 Accuracy cannot certify mechanism

Two estimators with the same error on a dataset can compute different functions of their input. One routes its prediction through transit time. The other routes it through confounds that happen to vary with pressure in that dataset. The two diverge only under distribution shift, which is the condition of deployment. Verifying a model therefore requires intervening on it, not only evaluating its outputs.

5.3 Project 3A: A Mechanistic Audit for Blood Pressure Estimators

The audit asks whether an estimator’s prediction depends on transit time in the direction and magnitude that Equation (5.1) requires. It has three tiers of increasing strength.

1. Linear probes test whether PTT can be read linearly from the model’s internal activations.

2. Nonlinear probes test whether it can be read at all, separating absence from nonlinear encoding.
3. Causal tests intervene on the model. PTT-direction ablation removes the decoded direction and measures the change in prediction. Donor-swap activation patching transplants activations from an input with a different transit time and checks that the prediction moves by the amount the physics predicts. A roll test delays the distal PPG, which lengthens PTT, so the predicted pressure must fall. Each effect must have the correct sign and a physically plausible size.

A model that is accurate but fails every tier is classified as not encoding transit time. At run time such a model abstains rather than reporting a pressure it cannot justify.

Synthetic testbed. Real data cannot validate an audit, because the true mechanism of a trained network is unknown. We therefore built a testbed in which it is known. Part 1 is a scalar task solved by three small networks with different, known mechanisms. Part 2 is a time-domain tube-load waveform simulator in which foot-to-foot PTT was verified to be unchanged by the load parameter Γ .

Preliminary result. Across the Part 2 sweep, the donor-swap effect increased steadily with α , the degree to which the model relies on transit time, while accuracy stayed flat. Accuracy did not change and the audit did, which is direct support for the argument that accuracy cannot certify mechanism.

Remaining work. Part 3 applies the audit to published architectures trained on real ECG and PPG. Whether most such models fail the audit is the hypothesis of that experiment, not yet a result. We also piloted a camera-based measurement of transit time between the face and the hand. The remote PPG signals were too noisy to give reliable transit times, and camera-based systems of this kind have already been reported [3], so this route is not pursued.

5.4 Project 3B: Hemodynamic Attenuation of the Micro Architecture

An audit is only practical when the computation it examines is small. In Chapter 4, a trained transformer carried its causal computation in two of forty heads, but this was found only after training, by ablating each head in turn, and the number of heads to keep had to be chosen by hand. Project 3B asks whether a model can find the number of heads it needs during training, without being given it. We introduce a rule for this, hemodynamic attenuation, and test it on tasks whose minimal number of heads is known.

5.4.1 Motivation from cerebral blood supply

Four properties of cerebral blood supply motivate the rule:

1. Supply follows need. Local blood flow rises with neural activity, signalled in part by astrocytes [5]. Each head is given a valve $g_h \in [0, 1]$.

2. A vessel is dispensable when collateral vessels can supply its tissue [6]. A head is valued by what is lost when it closes and the other heads compensate, which we call its collateral value.
3. Maintaining a vessel has a metabolic cost [7]. Each open head has a fixed price, and there is no target number of heads.
4. Unneeded vessels regress gradually [8, 9], and collaterals can enlarge when a route is blocked [6]. Closing heads fade out over 100 training steps and can be reopened.

The biology motivates the design, and no claim is made about the brain.

5.4.2 Astrocytes and attention

Softmax attention turns a query’s scores $s_1, \dots, s_N$ into weights

$$a_i = \frac{e^{s_i}}{\sum_j e^{s_j}}, \quad (5.3)$$

a numerator that differs for each memory slot and a denominator that is one number shared by all of them. Kozachkov et al. [4] showed that a network of neurons and one astrocyte can approximately compute self-attention. The neurons store the numerators in Hebbian synapses. The astrocyte computes the denominator. It pools activity across all the synapses it wraps and scales every one of them by the same gain, $1/p_t$. For token t the output is approximately (their Eq. 11)

$$\ell_t \approx \frac{1}{p_t} \sum_i e^{k_i^\top q_t} v_i + x_t, \quad p_t = \sum_j e^{k_j^\top q_t}, \quad (5.4)$$

which is softmax attention plus a residual connection. The point used here is that the astrocyte acts as a single gain on a whole block of synapses.

5.4.3 Head pruning

Michel et al. [13] remove heads by a gradient score, and Voita et al. [14] learn head gates with an L_0 penalty. Optimal Brain Damage [10] scores a parameter by the loss increase when it is removed, and Optimal Brain Surgeon [11] by the increase after the remaining parameters are re-fitted. ZipLM [12] applies the latter to whole heads of trained language models. All of these prune after training, to a size chosen by the user.

5.4.4 Task

The memory has $N = 16$ slots, each holding a random item c_j . A query specifies a position p , and the target is the items at R fixed offsets (Figure 5.1a):

$$\text{target}(p) = (c_{p+\delta_1}, \dots, c_{p+\delta_R}), \quad \text{slots modulo } N. \quad (5.5)$$

Items are redrawn for every example, so the model must learn where to attend.

A head returns, for each query, one weighted average of the items, which is one linear equation in the R unknown items. Recovering R items needs R independent equations, so the task needs $k^* = R$ heads, and with $k < R$ heads the lowest achievable loss is

$$\text{loss}(k) = \left(1 - \frac{k}{R}\right) L_{\text{triv}}, \quad (5.6)$$

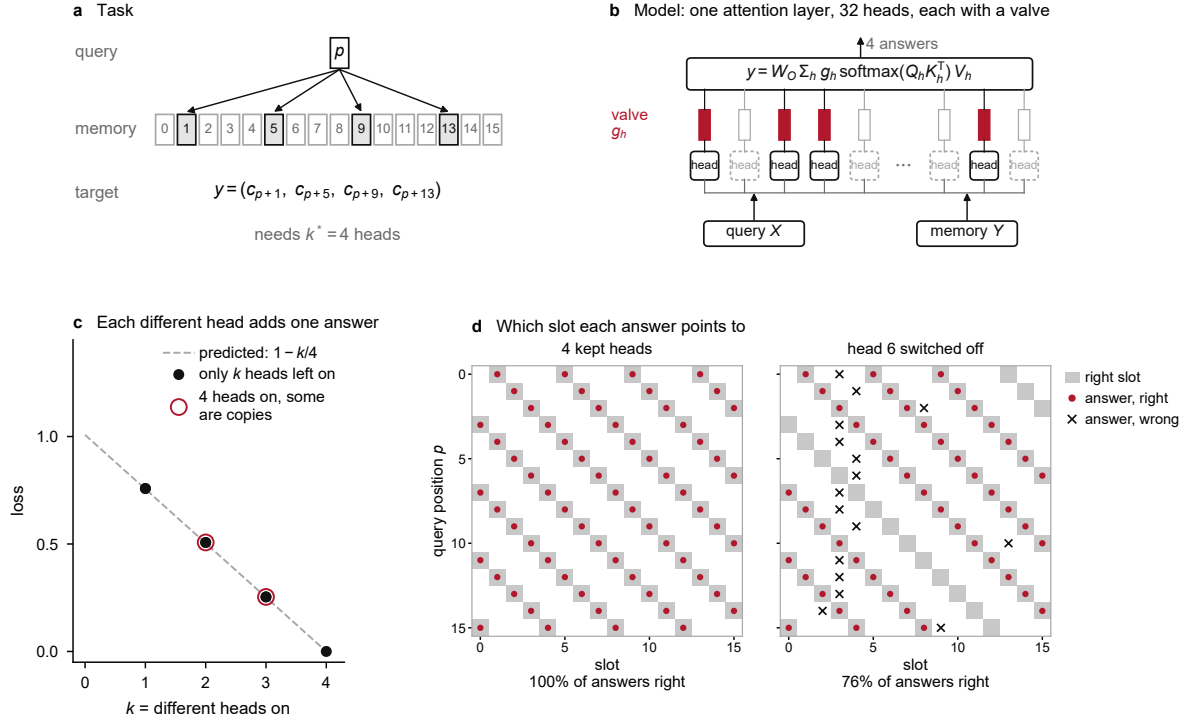


Figure 5.1: Task and model. **a**, A query carries a position p , and the target is the items stored 1, 5, 9, 13 slots ahead, so 4 heads are needed. **b**, One attention layer with 32 heads, each with a valve g_h . Filled valves are open. **c**, Loss with k different heads on a trained 4-head model, against the prediction $1 - k/4$. **d**, The slot each answer points to. All are correct with the 4 kept heads, and 76% with head 6 switched off.

where L_{triv} is the loss of predicting the mean. Duplicate heads give the same equation and count once. Measured losses agree with Equation (5.6) to within 1% (Figure 5.1c). The task counts as solved when the loss is below $0.02 L_{\text{triv}}$. Results are unchanged for any threshold from 0.01 to 0.1.

5.4.5 Model

The model is one cross-attention layer with $H = 32$ heads and no MLP. Matrices have one column per token, as in [4], with X holding the queries and Y the memory. Head h computes the usual attention,

$$O_h = V_h \text{softmax}\left(\frac{K_h^\top Q_h}{\sqrt{d_k}}\right), \quad Q_h = W_Q^h X, \quad K_h = W_K^h Y, \quad V_h = W_V^h Y, \quad (5.7)$$

with the softmax taken down each column, so each query’s weights over the memory add up to one. Each head’s output is multiplied by its valve g_h before the heads are added together:

$$\hat{T} = \sum_{h=1}^H g_h W_O^h O_h. \quad (5.8)$$

A head with $g_h = 0$ contributes nothing and receives no gradient, so a model with k open valves is exactly a k -head model. The valve plays the same algebraic role as the astrocyte gain, one

number that multiplies a whole block of synapses, here the output weights W_O^h of one head. The two differ in what sets them and how fast they change (Table 5.1).

	Astrocyte gain [4]	Head valve
One number for	all synapses of one astrocyte	all output weights of one head
Set by	pooled input activity	collateral value relative to a price
Changes	with every token	slowly, over training
Function	softmax denominator	selection of heads

Table 5.1: Two gains on blocks of synapses. The correspondence is algebraic only.

5.4.6 Hemodynamic attenuation

All valves start open. After a warm-up of 10% of training, every 25 steps the rule (1) computes the collateral value of every open head, (2) closes the lowest-valued head if its value is below the price $0.03 L_{\text{triv}}$, and (3) reopens the highest-valued closed head if its value exceeds twice the price. A closing valve decreases by 1/100 per step.

The collateral value is the increase in the best-fit loss when head h is removed and the other open heads re-fit their output weights:

$$v_h = \min_W L(\text{without } h) - \min_W L(\text{all open heads}). \quad (5.9)$$

This is the Optimal Brain Surgeon score [11], applied to whole heads as in ZipLM [12], and it is computed in closed form from one matrix inverse. The score is not new. The contribution is its use during training, with a price, a gradual fade, and reopening. The re-fit is what makes a copy worthless. If two heads output the same signal with weights (0.5, 0.5), removing one without re-fitting halves the output, so both look important. With re-fitting the other head takes weight 1, the loss does not change, and the copy is worth nothing.

5.4.7 Baselines and evaluation

All methods share the model, task, training and runs. The published methods are run inside our training: ZipLM-style pruning [12] (the same score applied once, then fine-tuning), Michel et al. [13], and Voita et al. [14]. Three controls each remove one part of the rule: no fade, no re-fit (the Optimal Brain Damage score [10]), and no score (random choice). We report exact count (open heads over the last 40% of training equal k^*) and never broke (after first falling below the solved threshold, the loss never exceeds it), on $k^* = 2, 4, 8$ with 10 seeds each, together with runs with duplicated heads, runs with random head failure, and a two-layer induction task [15].

5.4.8 Preliminary results

Count and stability. Hemodynamic attenuation kept exactly k^* heads in 29 of 30 runs and never broke the model in 30 of 30. No other method achieved both (Table 5.2). Without the fade the model broke in every run. Without the re-fit it kept too many heads. With random choice it broke in 24 of 30 runs.

Method (30 runs each)	Exact count	Never broke
Hemodynamic attenuation	29/30	30/30
ZipLM-style [12]	24/30	0/30
Michel et al. [13]	28/30	0/30
Voita et al. [14]	9/30	0/30
No fade	30/30	0/30
No re-fit	2/30	30/30
No score (random)	30/30	6/30

Table 5.2: Exact count and stability on the retrieval task ($k^* = 2, 4, 8, 10$ seeds each).

Duplicated heads. With every head starting as a copy of another, the rule kept 4 heads and never both members of a pair. Without the re-fit, 8 to 15 heads and 4 to 7 pairs were kept (Figure 5.2d).

Backup heads. When heads fail at random, the rule should keep extra heads in proportion to how likely they are to be needed. A formula written before the experiments predicted the number kept exactly in 21 of 24 runs and to within one head in the rest (Figure 5.2e).

Induction. In a two-layer attention-only model with $8 + 8$ heads, the minimal induction circuit is one previous-token head and one induction head [15]. Hemodynamic attenuation kept $2 + 1$ heads in 10 of 10 runs and never broke the model. The kept heads had the expected roles in all 10 runs, and switching off the induction head reduced next-token accuracy on the repeat to zero (Figure 5.3). Layer 2 reached the needed single head, while layer 1 kept two. Every method kept the same $2 + 1$ heads on this task, so only stability separated them. The method of Michel et al. broke the model in all 10 runs.

Cost. The final loss was 2 to 7 times the dense model’s, well below the threshold, using 0.31 to 0.45 of its head-steps. The saving is nominal, since closed heads are still computed.

5.4.9 Proposed work

The rule currently values a head by re-fitting only the final read-out. Between layers, one layer’s heads change what the next layer attends to, and a re-fit at the output cannot repair that. This is why layer 1 of the induction model keeps one head more than needed. The proposed extension values a head at its own layer, by asking the other heads of that layer to rebuild what the next layer reads, the same layer-wise approach used to prune large language models [12]. The same extension is needed to apply the rule at scale. We will then test the rule on qSA-Cross, asking whether it keeps the two causal heads found in Chapter 4, and on the Cross-Site Transformer with WildPPG, asking whether the same heads are kept across all sixteen leave-one-subject-out folds.

5.5 Timeline

5.6 Risks and Fallbacks

The audit is inconclusive on real data. The audit’s contribution does not depend on new human data. If published models cannot be retrained or their activations are unavailable, the audit is reported on the synthetic testbed and on models trained here on public multi-site recordings.

The rule does not reach the minimum in deeper models. This is already visible in the induction task, where layer 1 keeps one extra head. If the layer-wise value does not close the gap, the result is still a small circuit found during training, and the gap is reported with the number of extra heads per layer.

The rule does not scale. The collateral value needs one matrix inverse per layer. If this is too costly for large models, it will be computed on a subset of layers or with the approximations used in large-model pruning, and the accuracy of the approximation will be measured on the small tasks where the exact value is known.

The kept heads differ from those found in Chapter 4. This would be a finding about how stable circuits are across training methods, and will be reported as such.

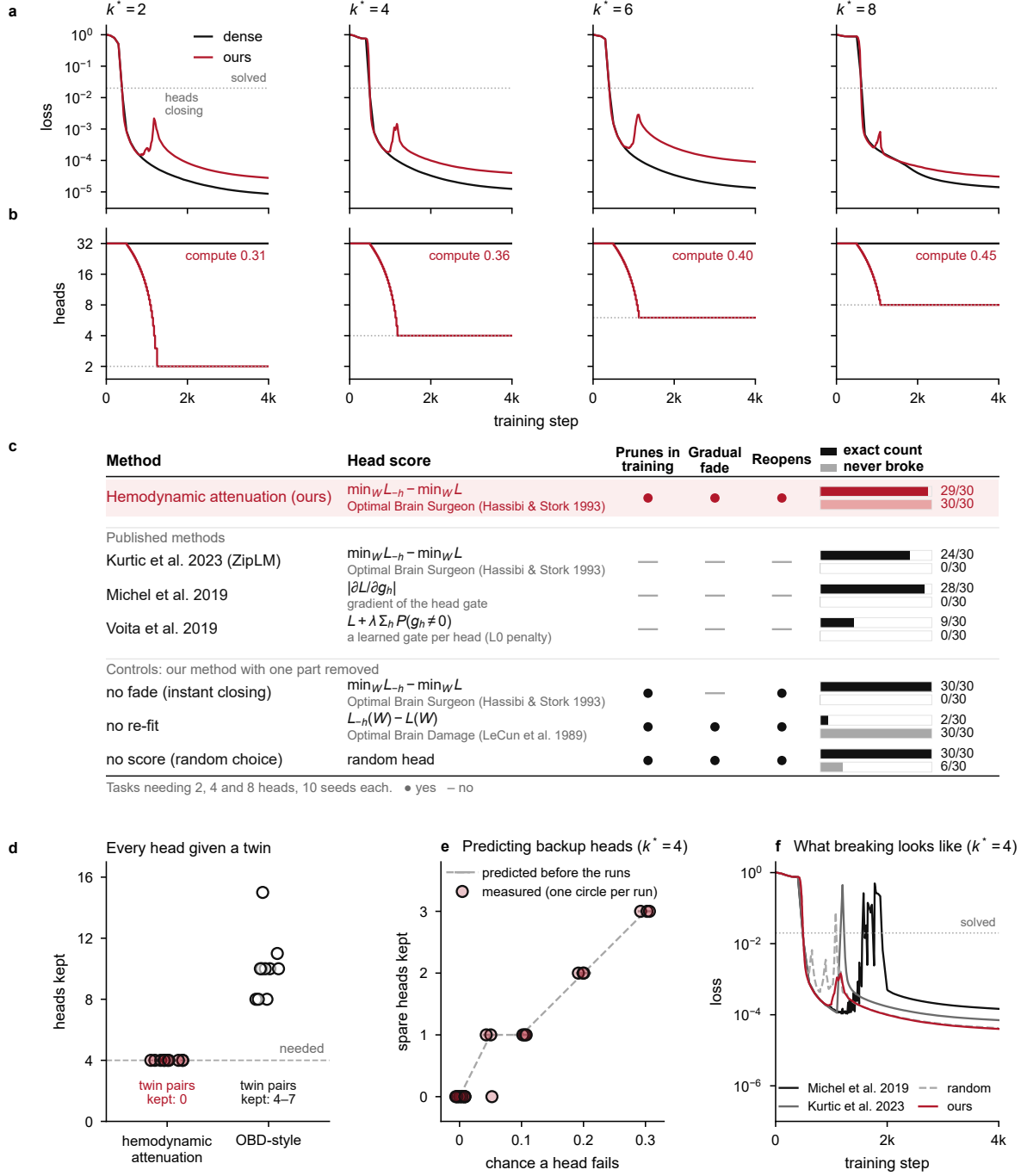


Figure 5.2: Hemodynamic attenuation during training. **a**, Loss of the dense model (black) and with hemodynamic attenuation (red). **b**, Open heads, with k^* dotted. **c**, All methods on the same 30 runs. **d**, Runs in which every head starts with an identical copy. **e**, Spare heads kept under random failure, against the prediction. **f**, One run per method. Only hemodynamic attenuation stays below the solved threshold.

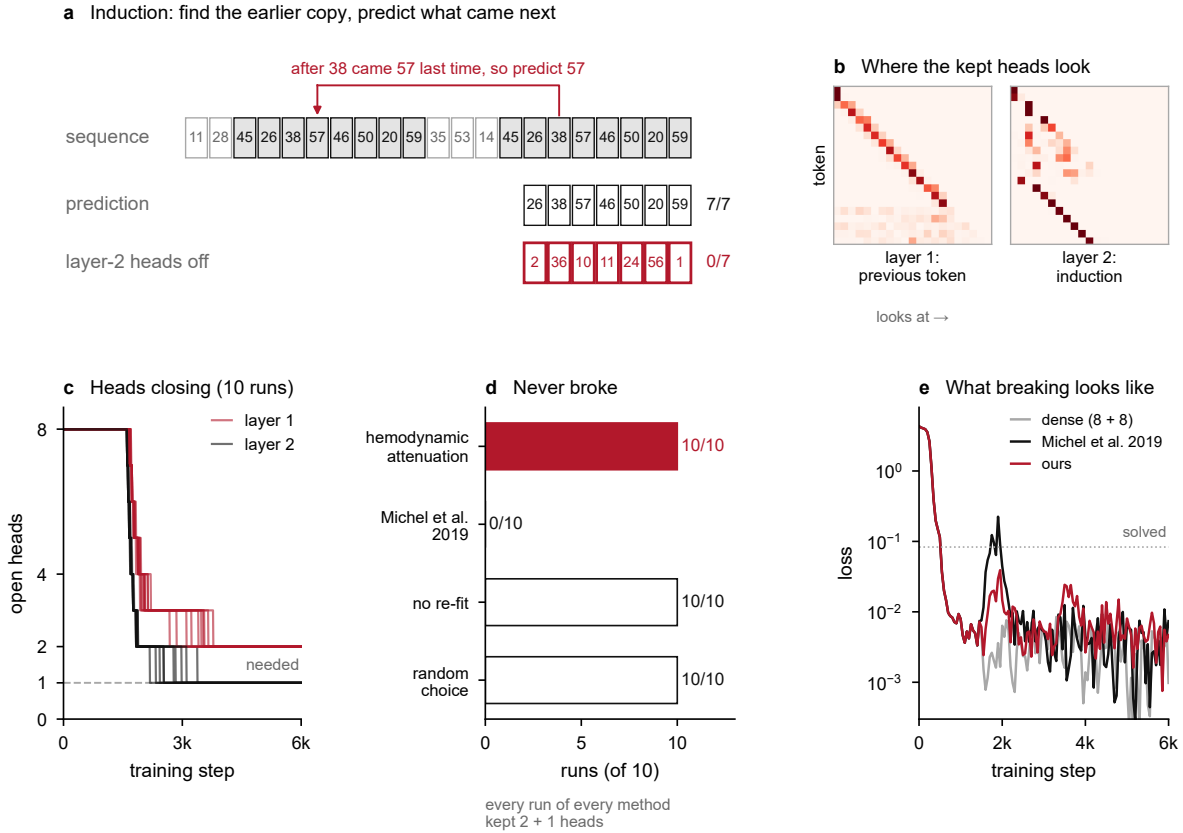


Figure 5.3: Two-layer induction. **a**, An example sequence and the model’s predictions on the repeat, with all kept heads (7/7 correct) and with the layer-2 head switched off (0/7). **b**, Attention of the kept heads on that sequence. The layer-1 head looks one token back, and the layer-2 head looks at the token after the earlier copy. **c**, Open heads in each layer during training, 10 runs. Layer 2 reaches the needed one head, and layer 1 stops at two. **d**, Runs that never broke, 10 per method, all with the same settings. **e**, Loss during training for one seed.

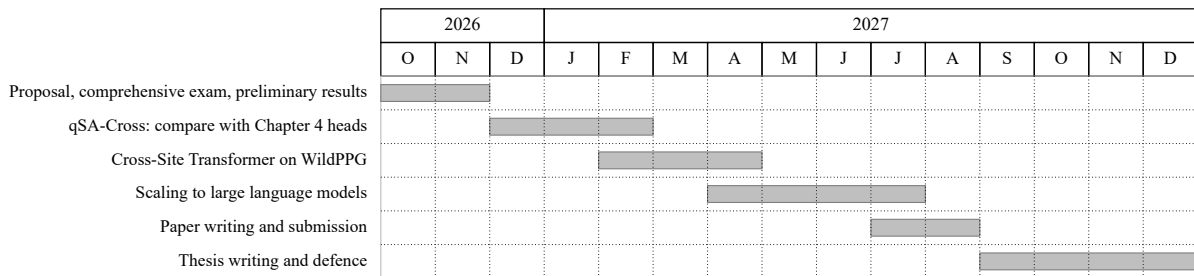


Figure 5.4: Timeline from the proposal to the defence.

Bibliography

- [1] D. J. Hughes, C. F. Babbs, L. A. Geddes, J. D. Bourland. Measurements of Young’s modulus of elasticity of the canine aorta with ultrasound. *Ultrasonic Imaging* 1(4):356–367, 1979.
- [2] R. Mukkamala et al. Toward ubiquitous blood pressure monitoring via pulse transit time: theory and practice. *IEEE Trans. Biomed. Eng.* 62(8):1879–1901, 2015.
- [3] I. C. Jeong, J. Finkelstein. Introducing contactless blood pressure assessment using a high speed video camera. *J. Med. Syst.* 40:77, 2016.
- [4] L. Kozachkov, K. V. Kastanenko, D. Krotov. Building transformers from neurons and astrocytes. *PNAS* 120(34):e2219150120, 2023.
- [5] D. Attwell, A. M. Buchan, S. Charpak, M. Lauritzen, B. A. MacVicar, E. A. Newman. Glial and neuronal control of brain blood flow. *Nature* 468:232–243, 2010.
- [6] J. E. Faber, W. M. Chilian, E. Deindl, N. van Royen, M. Simons. A brief etymology of the collateral circulation. *Arterioscler. Thromb. Vasc. Biol.* 34(9):1854–1859, 2014.
- [7] C. D. Murray. The physiological principle of minimum work: I. The vascular system and the cost of blood volume. *PNAS* 12(3):207–214, 1926.
- [8] Q. Chen et al. Haemodynamics-driven developmental pruning of brain vasculature in zebrafish. *PLoS Biol.* 10(8):e1001374, 2012.
- [9] C. A. Franco et al. Dynamic endothelial cell rearrangements drive developmental vessel regression. *PLoS Biol.* 13(4):e1002125, 2015.
- [10] Y. LeCun, J. S. Denker, S. A. Solla. Optimal brain damage. *NeurIPS* 2, 1989.
- [11] B. Hassibi, D. G. Stork, G. J. Wolff. Optimal brain surgeon and general network pruning. *IEEE International Conference on Neural Networks*, 1993.
- [12] E. Kurtic, E. Frantar, D. Alistarh. ZipLM: Inference-aware structured pruning of language models. *NeurIPS*, 2023.
- [13] P. Michel, O. Levy, G. Neubig. Are sixteen heads really better than one? *NeurIPS*, 2019.
- [14] E. Voita, D. Talbot, F. Moiseev, R. Sennrich, I. Titov. Analyzing multi-head self-attention: specialized heads do the heavy lifting, the rest can be pruned. *ACL*, 2019.
- [15] C. Olsson et al. In-context learning and induction heads. *Transformer Circuits Thread*, 2022.